

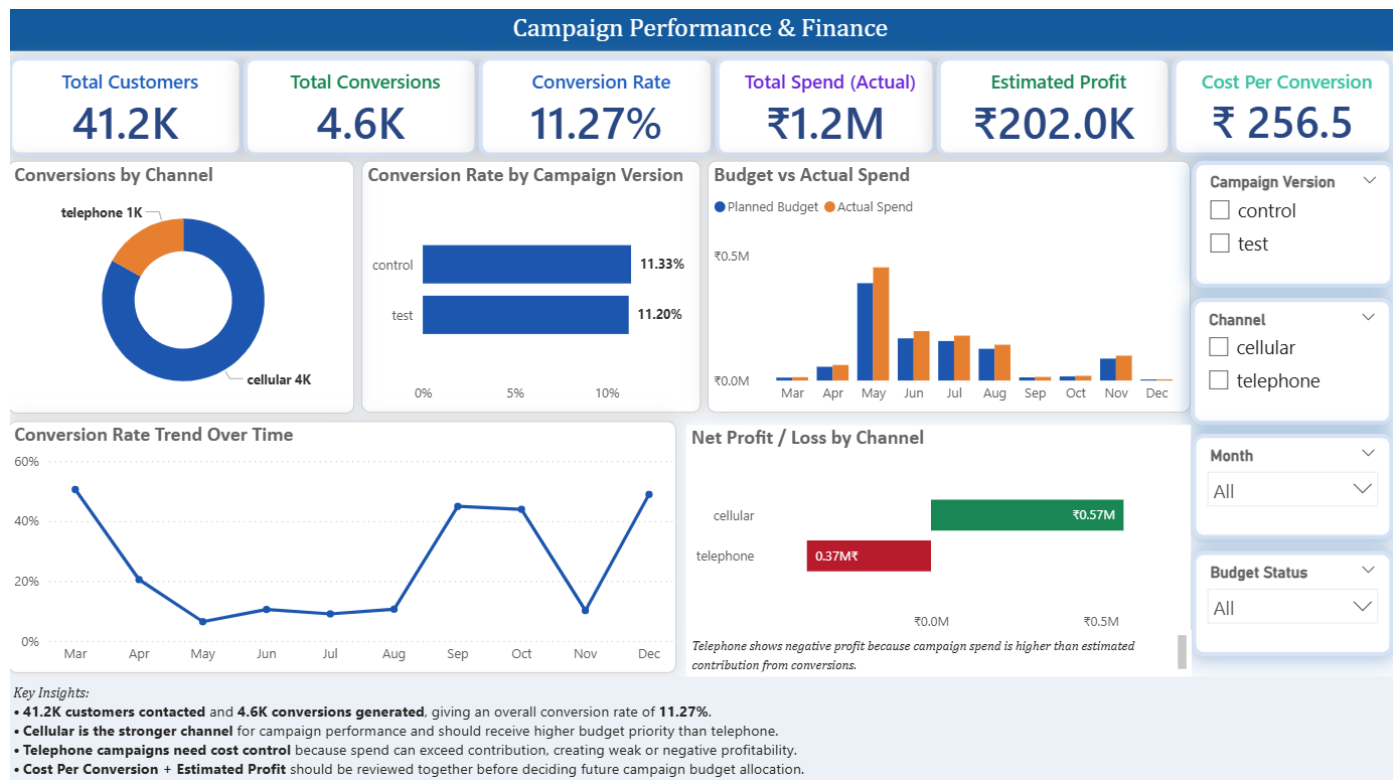
BFSI Campaign Analytics Project

AI-Assisted Bank Term Deposit Campaign Optimization | Python • SQL Server • Power BI

Professional Data Analytics Case Study Report

Customers Contacted	Conversions	Conversion Rate	Actual Spend	Estimated Profit	Cost / Conversion
41.2K	4.6K	11.27%	₹1.2M	₹202.0K	₹256.5

Project objective: Improve term deposit campaign performance by combining customer segmentation, campaign testing, cost control, lead scoring, and executive dashboarding.



Dashboard Page 1: Campaign Performance & Finance

1. Executive Summary

- Built an end-to-end BFSI analytics project around bank term deposit campaign performance using Python, SQL Server, Power BI, DAX, A/B testing, and lead scoring.
- Started with the UCI Bank Marketing dataset, which represents phone-based direct marketing campaigns for a Portuguese banking institution where the target is term deposit subscription.
- Created a connected analytics model with customer, campaign, experiment, campaign cost, A/B result, and ML lead score outputs.

- Delivered two Power BI dashboard pages: Campaign Performance & Finance and Customer Targeting & Lead Scoring.
- Used AI assistance only for selected workflow support: business framing, code/debug review, and insight wording; all calculations were executed and validated manually.

2. Business Problem

A bank wants to improve term deposit campaign performance. Management needs to know which campaign version works, which channel is more efficient, which customers are most likely to convert, and whether campaign spending is justified by estimated profit. The project turns raw campaign data into management-ready insights and recommended actions.

Stakeholder	Question Answered	Dashboard Area
Marketing Manager	Which campaign version/channel converts better?	Campaign performance visuals
Finance / FP&A Team	Is spend controlled and profitable?	Spend, profit, cost per conversion
CRM / Sales Team	Which customers should be prioritized?	Lead score bands and top high-probability customers
Leadership	What should be scaled, reduced, or retested?	Key insights and recommendations

3. Data Sources and Assumptions

Data Asset	Source / Creation	Business Purpose
bank-additional-full.csv	UCI Bank Marketing dataset	Customer profile, campaign contacts, conversion outcome
campaign_cost_assumptions.xlsx	Created as a finance assumption file	Planned cost, actual cost, contribution assumptions by channel/version
Synthetic A/B assignment	Generated from bank customers	Allows connected control vs test comparison using shared customer_id and campaign_id
ML lead score output	Generated using Python / scikit-learn	Customer-level probability and lead score band

Important transparency note: The original public dataset does not include a real customer ID, campaign ID, campaign cost, or A/B variant. These were created as documented portfolio assumptions to demonstrate a realistic analytics workflow without pretending they were original fields.

4. End-to-End Analytics Workflow

Stage	What was done	Output
Data collection	Downloaded UCI banking data and created campaign cost assumptions	Raw CSV + Excel input files
Python cleaning	Standardized column names, trimmed text, removed duplicates, created customer_id and conversion flag	clean_bank_data.csv
Feature engineering	Created age groups, campaign pressure, previous contact flag, month number,	Cleaned analytical features

	and quality flags	
Experiment layer	Generated control/test version and campaign_id by channel/version/month	fact_experiment + dim_campaign
Finance layer	Calculated planned budget, actual spend, contribution, profit, variance, and budget status	fact_campaign_cost
A/B testing	Compared control vs test conversion rates using two-proportion z-test	ab_test_result.csv
ML lead scoring	Trained explainable logistic regression and exported probabilities/bands	ml_scores.csv
SQL Server	Loaded clean files into a local database and validated business metrics	BI-ready tables
Power BI	Created data model, DAX measures, visuals, slicers, and insight sections	Two-page dashboard

5. Data Model and SQL Layer

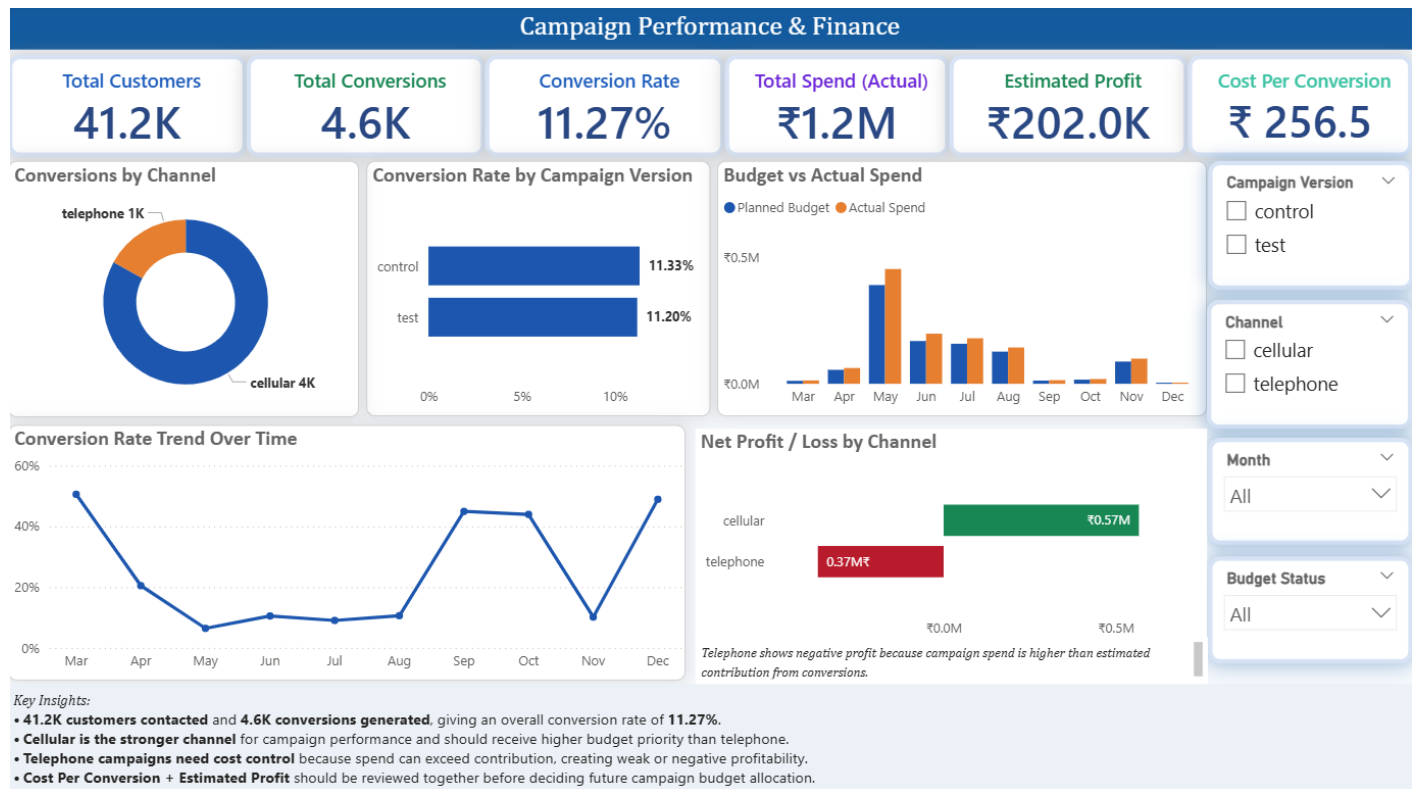
Table	Type	Purpose
dim_customer	Dimension	Customer demographics and profile attributes
dim_campaign	Dimension	Campaign ID, channel, version, month and month number
fact_experiment	Fact	Customer-level campaign assignment and conversion result
fact_campaign_cost	Fact	Budget, spend, contribution, profit, variance, and budget status
ml_scores	Output / fact-like table	Customer-level conversion probability and lead score band
ab_test_result	Summary output	Control/test conversion rates, lift, p-value, and recommendation

Relationship	Cardinality
dim_customer[customer_id] -> fact_experiment[customer_id]	One-to-many
dim_campaign[campaign_id] -> fact_experiment[campaign_id]	One-to-many
dim_campaign[campaign_id] -> fact_campaign_cost[campaign_id]	One-to-many
dim_customer[customer_id] -> ml_scores[customer_id]	One-to-one

6. Analytical Methods

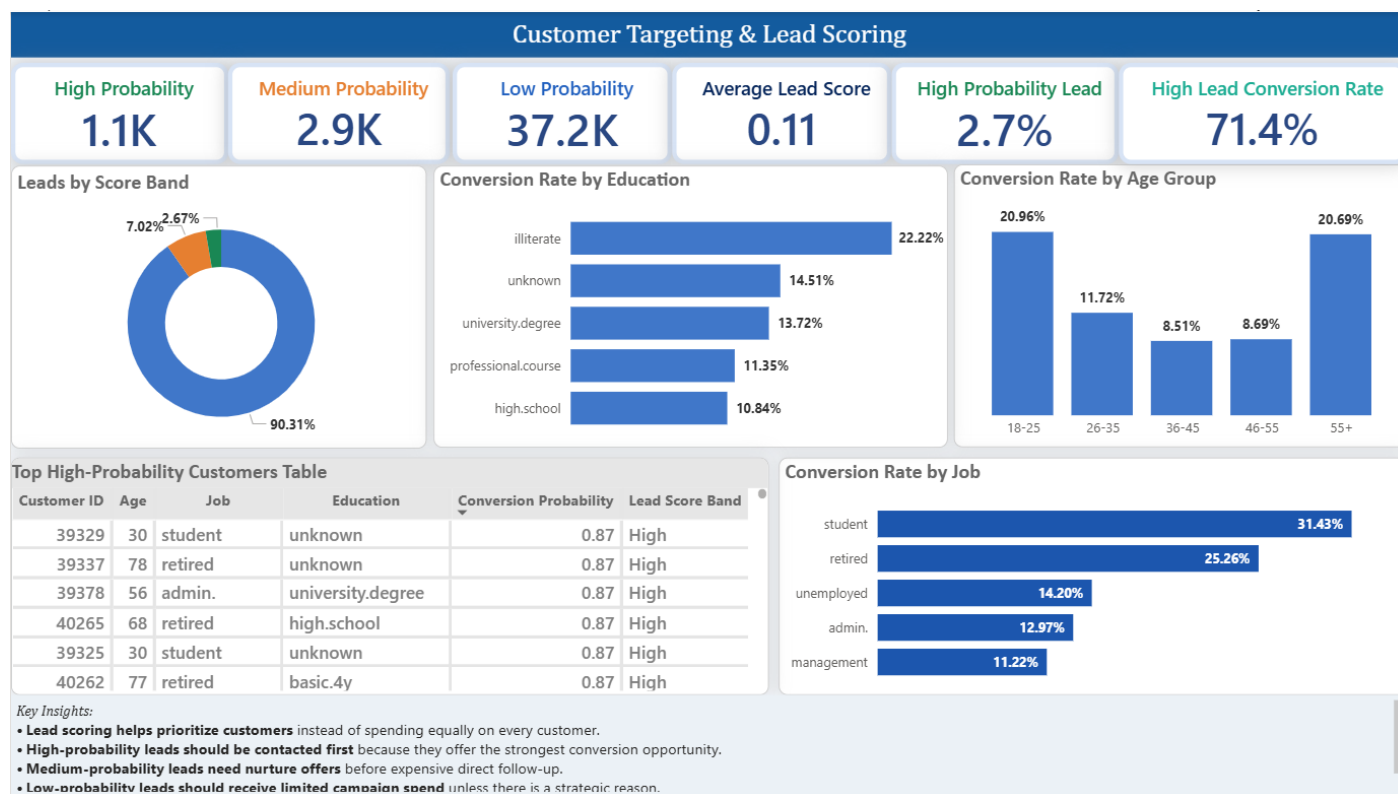
Method	Why it was used	Business Decision Supported
EDA	Understand conversion patterns by channel, age, job, education, and campaign version	Identify high-response customer segments
Budget variance analysis	Compare planned vs actual campaign spend	Control overspending and improve budget allocation
A/B testing	Compare control vs test campaign conversion rates	Decide whether to scale, stop, or retest the test version
Lead scoring model	Estimate customer conversion probability	Prioritize high-probability customers for follow-up

7. Dashboard Page 1: Campaign Performance & Finance



- The campaign contacted 41.2K customers and generated 4.6K conversions, resulting in an overall conversion rate of 11.27%.
- Cellular is the stronger channel, contributing most conversions and positive profitability compared with telephone.
- Telephone campaigns require cost control because campaign spend is higher than estimated contribution, creating a negative profit impact.
- Control and test campaigns perform almost equally, so the test campaign should be redesigned or retested before scaling.
- Cost Per Conversion and Estimated Profit should be reviewed together before future budget allocation decisions.

8. Dashboard Page 2: Customer Targeting & Lead Scoring



- Lead scoring separates customers by conversion potential, helping the bank prioritize outreach instead of contacting everyone equally.
- High-probability leads should be contacted first because they represent the strongest opportunity for conversion-focused follow-up.
- Medium-probability leads need nurture campaigns or personalized offers before high-cost direct calling.
- Low-probability leads should receive limited campaign spend unless there is a strategic business reason.
- Students, retired customers, and younger/older age groups show stronger response patterns, making them useful segments for targeted messaging.

9. AI-Assisted Workflow

AI-assisted area	How AI helped	Human validation
Business framing	Helped refine business problem, stakeholders, and KPI questions	Final KPIs selected manually based on dashboard purpose
Code/debug support	Helped review Python, SQL, and DAX logic when errors appeared	Code was executed in Jupyter, SQL Server, and Power BI
Insight wording	Helped turn technical outputs into concise executive bullets	Final wording was edited and matched to actual dashboard visuals

Positioning: AI was used as a support tool, not as a replacement for analysis. The data cleaning, calculations, SQL loading, dashboard build, and validation were performed by the analyst.

10. Limitations and Next Steps

- The A/B campaign version is synthetic because the public dataset does not include real randomized campaign variants.
- Campaign cost and contribution assumptions were created for portfolio demonstration and should be replaced with actual finance data in a production setting.
- The lead scoring model is a baseline logistic regression model; future improvement could include model comparison, calibration, and lift charts.
- Next version could add automated refresh, SQL views, a model documentation page, and a README case study for GitHub.

11. Interview Story

Suggested explanation: “I built an end-to-end BFSI campaign analytics project using a real public banking dataset. I cleaned and transformed the data in Python, created a connected experiment and campaign-cost model, loaded clean tables into SQL Server, performed A/B testing and ML lead scoring, and built a two-page Power BI dashboard. The dashboard helps management understand conversion, cost efficiency, channel profitability, and which customers should be prioritized for future outreach.”

12. References

UCI Machine Learning Repository - Bank Marketing Dataset:

<https://archive.ics.uci.edu/dataset/222/bank%2Bmarketing>

Microsoft Learn - Power Query: <https://learn.microsoft.com/en-us/power-query/power-query-what-is-power-query>

Microsoft Learn - Power BI Star Schema Guidance: <https://learn.microsoft.com/en-us/power-bi/guidance/star-schema>

statsmodels - proportions_ztest documentation:

https://www.statsmodels.org/dev/generated/statsmodels.stats.proportion.proportions_ztest.html